From Video to Rig: Markerless Horse Motion Capture for Animation

Keypoints, an anatomical solver and a rig-aware retargeting step, measured against optical mocap

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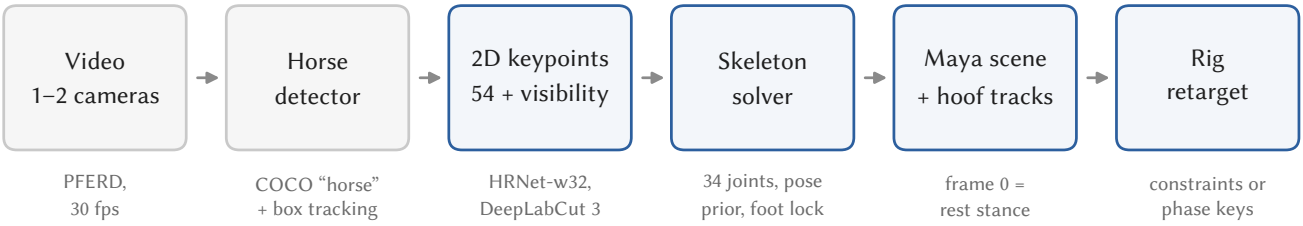


Figure 1: Pipeline. Blue boxes are the parts built for this work; the horse detector is an off-the-shelf COCO model with our box tracking. Every stage runs locally on one workstation (RTX 4080); the Maya scene is a plain `.ma` file and the rig step works on an unmodified production rig.

ABSTRACT

We describe a markerless pipeline that turns video of a horse, from one camera or a calibrated and synchronised pair, into animation on a production horse rig in Autodesk Maya. A 54-point keypoint network, trained on the PFERD and Horse-30 datasets, predicts body landmarks together with their visibility. An anatomical 34-joint solver combines a full-covariance pose prior, per-leg re-initialisation, video-driven foot locking, an optional second camera and a carpus hyperextension limit. A retargeting tool then drives the rig's IK legs through live constraints, or bakes keys on frames shared by all controls of a leg and placed at the phases of the step, with planted hooves holding the heading they had at touchdown. On an 80-second trial with synchronised optical motion capture, the two-camera configuration reaches a mean leg-flexion error of 6.2°, a pelvis error of 1.7 cm on the floor plane and 1.0 cm in depth, and a median hoof slide of 2.3 cm per stance (1.4 cm for the mocap itself); a single camera remains limited along its viewing ray. We report an ablation over cameras, priors and constraints, the generalisation of the keypoint network to horses it has never seen, and feedback from a production animator.

1 INTRODUCTION

Quadruped animation remains expensive. Optical capture of a horse needs a stage, markers glued to the coat and a handler, and hand-keyed locomotion takes an animator days per shot. Markerless capture for humans is now routine (e.g. Pose2Sim [9]); for horses, animal pose estimation has matured on the 2D side [7, 11], but the path from 2D points to animation an animator can use on an existing rig is rarely described end to end.

This work started as a markerless human capture tool and was extended to horses. Its goal is practical: the output is judged in Maya, on a production rig, by animators. That goal shapes the design. The solver must return a clean joint hierarchy rather than point clouds; feet must stay planted; the rig must not flip its legs; and keys, when baked, must be editable. We make four contributions:

- a 54-point horse keypoint network that also learns *visibility*, evaluated on unseen horses;
- an anatomical solver whose main components (full-covariance pose prior, foot locking, second camera, carpus limit) are measured against optical mocap in one ablation;
- a rig-aware retargeting step: body-anchored knee targets, planted hooves that hold their touchdown heading, and key reduction on frames shared by all controls of a leg;
- an evaluation against synchronised optical motion capture and feedback from a production animator.

2 RELATED WORK

Animal keypoints. DeepLabCut [4, 7] made transfer learning for animal keypoints standard; SuperAnimal [11] provides foundation models for quadrupeds, which we use as initialisation. Our backbone is HRNet [10]. Horse-10 [8] (released with 30 horses) studies out-of-domain robustness across individuals; PFERD [6] provides multi-view video of five horses with synchronised marker-based motion capture, which makes quantitative 3D evaluation possible.

Model-based animal capture. SMAL [12] and its horse variant hSMAL [5] fit a parametric shape and pose model to images. We instead fit a rig-like joint hierarchy with fixed bone lengths, which maps directly onto animation rigs. Fitting a kinematic model to 2D keypoints with priors follows SMPLify [2]; joint-limit priors in the spirit of [1] motivate our per-joint and full-covariance terms.

Clean-up for animation. Foot skate removal is a classic problem [3]; we handle it inside the solver (pins from video-detected contacts) and again on the rig (heading-held planted hooves).

Table 1: Keypoints on a horse unseen in training (PFERD horse 5, 6,000 frames). *Found*: confidence above the working threshold of each version (v1: 0.5, v2: 0.2; v2's confidences are calibrated lower). *Hit*: found and closer than 9.4 cm (a quarter of the radius length).

	Found (%)		Hit (%)	
Body region	v1	v2	v1	v2
Lower legs	95.7	**99.8**	90.3	**95.0**
Upper legs, pelvis	62.8	**89.0**	59.7	**83.9**
Back, tail	89.5	**99.5**	83.3	**90.2**
Neck	64.0	**97.8**	62.5	**95.7**
Head	80.6	**98.9**	75.0	**92.6**

Median position error of v2: 2.2 cm (lower legs), 3.1 cm (upper legs). Left/right confusions on the legs: 0.9 % (lower), 0.3 % (upper).

3 DATA

PFERD [6] contains five horses filmed by ten synchronised 1920×1088 video cameras (mostly 30 fps) with optical motion capture at 240 Hz and calibrated cameras. We define 54 keypoints on PFERD markers, using the rear hoof marker as the hoof point, and obtain 2D labels by projecting the markers into every camera. Training frames are taken when the horse has moved more than 5 cm, at most every 0.33 s and at most 150 per trial, from all ten cameras: 44,139 images of horses 1–4 (all trials but the last of each). Points on the far side of the body are labelled as hidden rather than left empty. We add Horse-30 [8]: 11 of its 22 points are mapped onto ours, its images are upsampled to the source resolution and repeated three times (12,150 training images). Horse 5 (6,000 frames) and ten Horse-30 individuals are held out for testing.

4 2D KEYPOINTS AND VISIBILITY

We fine-tune an HRNet-w32 in DeepLabCut 3, initialised from SuperAnimal-Quadruped (39 points) with a new 54-point head. Version 2 differs from version 1 in that hidden points are trained as explicit "not visible" targets (background weight 1), mirror augmentation swaps left and right labels, colour and box shift/scale augmentation is added, and Horse-30 is mixed in. Training runs 20 epochs (batch 16, AdamW, learning rate $10^{-4} \to 10^{-5} \to 10^{-6}$) in 6.7 h. For version 1, generating targets on the GPU with cuDNN autotuning and eight data workers raised throughput from 1.46 to 2.69 iterations per second, and sixteen workers to 3.78, with bit-identical outputs. A COCO-trained detector restricted to the *horse* class, with box tracking, crops the animal; detectors fine-tuned on PFERD latched onto shadows outdoors and SuperAnimal's detector onto people.

Table 1 reports the held-out horse. Visibility is the main gain: the confidence of v2 separates visible from hidden points with an AUC of 0.993 (v1: 0.709), so the solver can trust a point's weight. On ten unseen Horse-30 individuals v2 finds 99.6 % of points (v1: 66.1 %) with an error of 1.43 % of the horse's height (v1: 3.58 %). Already version 1 lowered the median error on the 18 points shared with the original SuperAnimal model from 6.4 to 2.8 cm. All numbers in this section use crops from the labelled keypoints, so the detector is not involved.

5 SKELETON SOLVER

Model. The skeleton has 34 joints (28 of them rotate), with bone lengths and rest stance taken as medians over twelve mocap trials of horse 1 in the body frame, symmetric left and right. A pose $\boldsymbol{x}$ holds the root position and 28 rotation vectors (87 parameters); forward kinematics places the 54 landmarks on the bones.

Objective. Each frame minimises, with a soft-L_1 loss ρ and warm start from the previous frame,

$$E(\boldsymbol{x}) = \sum_{c,k} \rho\!\left(\frac{w_{ck}\,\|\pi_c(M_k(\boldsymbol{x})) - u_{ck}\|}{\sigma_c}\right) + \lambda_p \|W(\boldsymbol{\theta} - \boldsymbol{\mu})\|^2 + \lambda_t \|\boldsymbol{x} - \boldsymbol{x}_{t-1}\|^2 + \sum_{h \in \mathcal{S}_t} \frac{\|p_h(\boldsymbol{x}) - \bar{p}_h\|^2}{\sigma_f^2} + \sum_{s \in \{l,r\}} \frac{\max(0,\, \theta_s^{\min} - \theta_s)^2}{\sigma_k^2}, \quad (1)$$

where u_{ck} is keypoint k in camera c, w_{ck} ramps from 0 to 1 between confidence 0.1 and 0.4, and σ_c scales with the horse's image height. The terms are explained below.

Full-covariance pose prior. With per-joint limits alone, one camera produced a "limping" horse at trot: 2D points were right, depth was not. We therefore whiten all joint rotations together: $\boldsymbol{\mu}$ and W come from 6,561 frames of 3D solves of eleven trials of the same horse, excluding the evaluation trial. Joints are now coupled (at trot elbow, carpus and fetlock flex together), and the weight $\lambda_p = 2$ was chosen on three trot clips and one walk clip.

Leg re-initialisation. A leg can get stuck in a wrong bend for seconds. When its confident points lie more than 4σ from the projected skeleton, that leg is re-solved from the mean pose and from the mirrored partner leg, and the better start is kept. In development, this step together with dropping a floor-contact term cut the frames with leg points missed by more than 10 px from 16 to 0.9 % (one camera); it is on in every row of the ablation.

Foot lock. A second pass pins hooves in stance. Stance $\mathcal{S}_t$ is detected in the video: the hoof point moves less than 0.5 % of the horse's image height per frame, is confident, lies within 12 cm of the lowest hoof, for at least four frames. The pin $\bar{p}_h$ is the median hoof position of the first pass over that stance, with $\sigma_f = 2$ cm. The pass makes the solve 1.8 times slower.

Second camera. With PFERD calibration, residuals from both cameras are summed; a camera that sees fewer than three points in a frame is ignored there. The ground plane then also comes from calibration rather than from the hooves.

Carpus limit. The animator reported knees bending backwards on touchdown (Section 8). A one-sided penalty stops the carpus from hyperextending beyond -5° ($\sigma_k = 1^\circ$).

Smoothing and floor. Rotations are filtered with a zero-phase second-order Butterworth filter. In an earlier one-camera test (camera 21386, foot lock), 3 Hz left the angle error almost unchanged against 8 Hz (9.5 vs 9.4°) while the 99th percentile of hoof acceleration dropped from 24 to 8.5 cm/frame^2, close to the mocap's 8.1; all results below use 3 Hz. With one camera the floor is placed so that the lowest hoof sits at its median mocap height of 0.9 cm.

Table 2: Ablation against optical mocap (trial 0007, 2,400 frames). A–C: one camera; D–F: two cameras (D is B plus the second camera); F is the final system. A uses per-joint limits instead of the full-covariance prior. Pelvis on floor: median horizontal distance to the mocap pelvis. Depth: median pelvis error along the main camera's ray. Hoof slide: median per mocap stance. Time: one BLAS thread per run on a 2-core test machine.

	Configuration	Leg error (°)	Frames >15° (%)	Pelvis on floor (cm)	Pelvis depth (cm)	Hoof slide (cm)	Carpus < −7° (%)	Time (s/frame)
A	One camera, per-joint limits	13.5	34.5	31.0	29.7	23.2	21.7	0.19
B	+ full-covariance pose prior	10.4	15.5	12.8	12.5	20.6	17.8	0.13
C	+ foot lock	10.2	14.7	12.0	11.7	8.9	18.3	0.24
D	Two cameras, pose prior	6.4	1.2	1.7	1.1	2.9	20.8	0.13
E	+ foot lock	6.4	0.7	1.7	1.0	2.2	21.0	0.23
F	+ carpus limit (final)	**6.2**	1.0	1.7	1.0	2.3	**2.1**	0.23
	Optical mocap	–	–	–	–	1.4	–	–

6 RETARGETING TO A PRODUCTION RIG

The solver writes a Maya ASCII scene: joints with keys, frame 0 as the rest stance. Since the second version it also writes one hidden locator per hoof whose rotation is the hoof heading and whose attributes mark stance and hoof height (Section 6.1). The target is an unmodified production horse rig (712 joints, FK and IK legs).

Placement and rest match. The mocap is scaled by the ratio of shoulder and hip heights and placed over the rig's pelvis. In frame 0 the long leg bones of the mocap are rotated by the shortest arc to point like the rig's bones; their children keep their world rotation, otherwise the parent's correction leaked into the pastern and the hooves bent.

Constraints. Rig controls follow the mocap through live constraints with offsets taken in the rest frame. The tool can leave them live (no keys), bake every frame, or bake keys on step phases (Section 6.2), which is the default of its window. IK feet are pinned at the mocap fetlock. Knee targets are not placed at the mocap knee: a nearly straight front leg has its carpus on the "wrong" side of the leg line in more than half of the frames, and a target there rotated the leg plane by 60–70° per frame (99th percentile). The target is instead the midpoint of the leg line plus an offset fixed in the pelvis frame, at least 30 cm from the line; measured on the solved skeleton with a 40 cm offset, the rotation drops to 5–9° per frame. The hoof is flat on the ground in stance and follows only the pastern's pitch in the air, blended by hoof height (4 to 12 cm).

6.1 Planted hooves keep their heading

In the first version a flat hoof took the pelvis heading. The horse in our trial turns its body by a median of 11° during one stance (up to 44°), so a planted hoof turned with the body. The hoof locators now hold the pelvis heading of the touchdown frame for the whole stance and blend smoothly to the next touchdown in the air (Figure 2). Their stance labels agree with the mocap's in 88–91 % of frames per hoof (Figure 6).

6.2 Keys on step phases

Mocap baked on every frame is hard to edit, and per-curve key reduction puts keys of the knee and the foot on different frames, which the animator singled out as the main problem. We bake every frame and then keep, for all controls of one leg, the *same* frames: touchdown, last frame of stance and highest point of the swing, read from the hoof locators. Body controls start from the

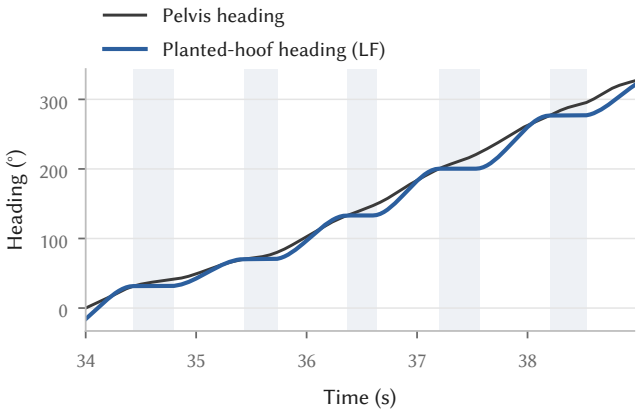


Figure 2: Heading during a tight turn of about 330° in 5 s. Shaded: stance of the left fore hoof. The planted hoof keeps its touchdown heading instead of turning with the pelvis.

Table 3: Key density after reduction (share of the 2,401 frames that carry a key, and mean gap in frames), computed on the IK targets of the final skeleton.

Tolerance (° / cm)	1	2	3	5
Legs, keyed frames (%)	40	29	24	18
Legs, mean gap (frames)	2.5	3.5	4.2	5.5
Body, keyed frames (%)	44	33	27	22
Body, mean gap (frames)	2.3	3.0	3.6	4.5

touchdowns of all legs. While a spline through the kept keys departs from the baked curve by more than a tolerance on any channel of the group, the frame of largest departure in each offending interval is added to the whole group. Planted IK feet get flat tangents. Figure 3 shows one leg; Table 3 shows key density against tolerance.

7 RESULTS

Protocol. We evaluate on PFERD horse 1, trial 0007: 2,400 frames (80 s at 30 fps) of walk, trot, turns and standing, with the horse 9–18 m from the cameras. The main camera is 23604; the second,

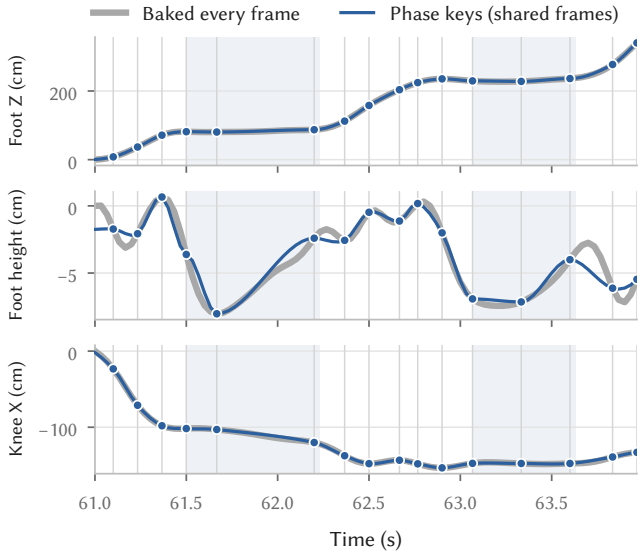


Figure 3: Keys on step phases, left fore leg, 3 s, tolerance 3 °/cm. Every channel of the leg is keyed on the same frames (vertical lines). Shaded: stance.

21386, sees the horse from a direction about 85° away (median). The reference is our skeleton fitted to the optical markers (median residual 0.94 cm). Leg error is the mean absolute difference of the flexion angle at twelve leg joints (elbow, carpus, fetlock, stifle, hock and hind fetlock on both sides). Hoof slide is the horizontal displacement of a hoof over one stance of the mocap. All configurations use the same 2D points and 3 Hz smoothing. The pose prior excludes this trial; the keypoint network and some parameters do not (Section 9).

Ablation. Table 2 isolates each component. The full-covariance prior is the largest gain for one camera (A→B): it couples joints so that the depth ambiguity of a single view no longer shows as limping. Foot lock (B→C, D→E) mainly reduces hoof slide, although one camera still slides 8.9 cm per stance (90th percentile 41.8 cm). The second camera removes most of the remaining depth error (C→E). The carpus limit (E→F) leaves the overall error almost unchanged while hyperextension beyond 7° becomes rare (Figure 7).

Viewpoint. Figure 4 splits the leg error by the horse's orientation to the main camera. With one camera the error grows when the horse walks towards or away from it, where leg swing runs along the viewing ray; the second camera flattens this dependence.

Motion quality. In the final system the pelvis height is 138.3 cm against 138.6 cm, the pelvis is 1.7 cm from the mocap on the floor plane (median), hooves move 8 cm/s in stance (mocap 5), and 1.0 % of frames have a mean leg error above 15°. Figure 5 shows the carpus and hock over the trial's best 8-second segment (leg error 4.5° there). On the RTX workstation the two-camera solve with foot lock took 519 s for the 80 s trial, 0.22 s per frame or 6.5 times slower than real time; the carpus limit adds under 1 %.

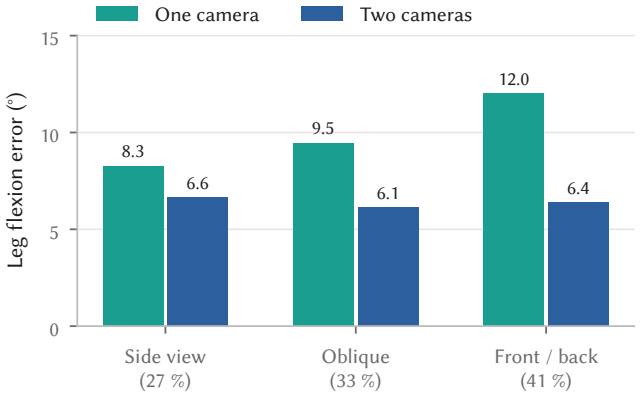


Figure 4: Leg error by orientation of the horse to the main camera (share of frames in brackets). Configurations C and E of Table 2.

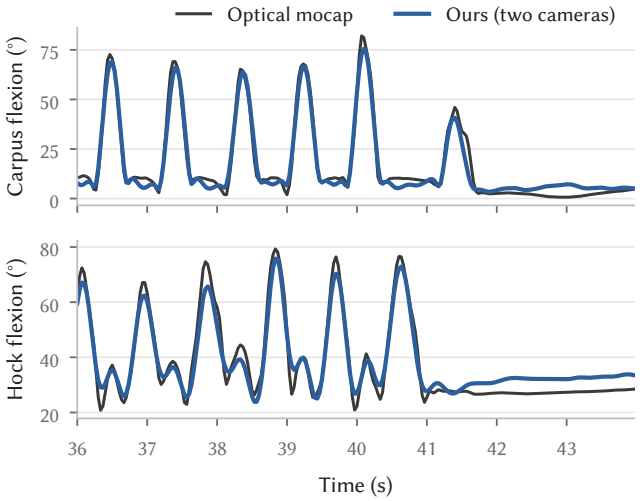


Figure 5: Flexion of the right carpus and hock, final system against optical mocap: the trial's best 8-second segment, 5.5 s of walk and trot followed by 2.5 s of standing.

8 ANIMATOR FEEDBACK

We showed a playblast of the rig next to the source video to an animator who had animated a horse in production. She reviewed the first version of the rig transfer. She confirmed that the footfall order matches the reference and found the hoof motion in the air lively. Her criticism, and what changed:

- *Knees bend backwards on touchdown and sometimes overextend in the swing, which reads as a jerk.* Measured about the body's lateral axis, the right carpus was hyperextended by 10° in stance against 4° in the mocap; with the limit it is 7°. About the joint's own axis, frames beyond 7° drop from 21.0 to 2.1 % (Table 2).
- *Jitter and sliding in stance.* Joint angles in stance jitter no more than in the mocap (2–3°), so we suspect the rig's IK near full extension; not yet verified.

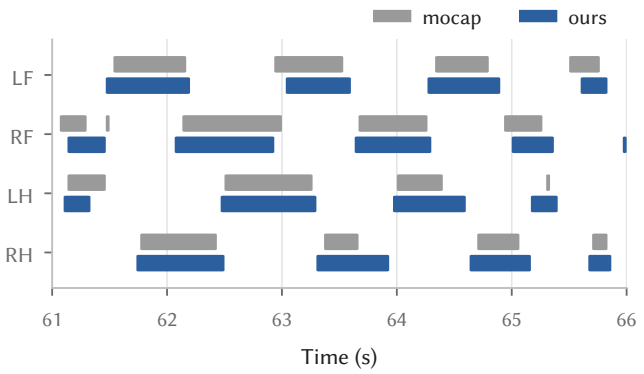


Figure 6: Footfall diagram for 5 s of walk: stance from the mocap (grey) and from our hoof locators (blue). LF, RF, LH, RH: left/right fore/hind.

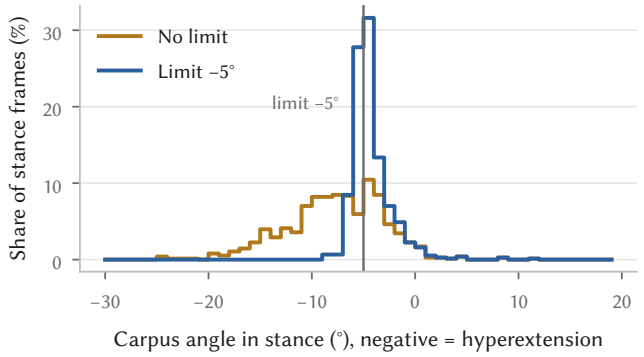


Figure 7: Right carpus angle in stance, measured about the joint's own bend axis, without and with the limit.

- *Legs slide and twist in tight turns.* The planted hoof followed the pelvis heading; now it holds its touchdown heading.
- *Every frame keyed, and keys of different controls on different frames.* She prefers keys on key poses. Answered by phase-aligned keys; live constraints remain available for those who key by hand.
- *The track drifts and shakes where the handler's legs hide the horse's legs in one view.* Not yet addressed.

She animates legs in IK so that feet stay planted, which matches our default. The heading hold and the phase keys have not yet been run in Maya or reviewed by her.

9 LIMITATIONS AND FUTURE WORK

The solver has been evaluated on one horse, whose proportions also define the skeleton. The keypoint network was evaluated on unseen horses, but its training set contains frames of the evaluation trial itself (from all ten cameras), so 2D accuracy on this trial is optimistic. The prior weight, the smoothing cut-off and the carpus penalty were chosen on segments of the same trial. Two cameras require calibration, which in our data only PFERD provides. With one camera, motion along the viewing ray remains the weak point. At trot the pose prior slightly reduces the range of leg swing. Occlusion by the handler is not handled; rejecting points on which the cameras disagree is the next step, as is checking IK near full extension on the rig. The rig stage of the first version has been judged visually in Maya; the heading hold and the key reduction have not yet run in Maya, and their numbers were computed offline on the IK targets.

10 CONCLUSION

With two calibrated, synchronised cameras, a horse can be captured with a mean leg-angle error of about 6° and the pelvis within 2 cm of optical mocap; a single camera is usable in side views (8.3°) but remains limited along its viewing ray. The measured gains came from the coupled pose prior and the second camera. On the rig side, body-anchored knee targets removed the leg flips, and the heading hold and phase keys answer the animator's remaining notes; they still need her review.

ACKNOWLEDGMENTS

PFERD is used under its non-commercial research licence; models trained on it are kept private. Horse-30 is distributed under CC BY-NC 4.0. We thank the animator who reviewed the results.

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